IWATANI CORPORATION · CONFIDENTIAL WORKING DRAFT

Share Reconciliation

Why India’s expansion joints share and bellows share differ, and a correction to the 5–10% figure

Client

Iwatani Corporation — Metals Department → Stainless Steel Division

Subject

Bellows tube (precision-slit material) market entry, leveraging the Jindal Stainless relationship

Status

Working document — updated at monthly internal meetings

Prepared

July 2026

Evidence standard. Every factual claim in this document carries an inline source link and a verification grade: **[P]** primary (company, association or government publication), **[C]** credible press, **[S]** commercial market-research vendor (indicative only), **[U]** unverified directory listing. Commercial estimates for the bellows market disagree with one another by up to a factor of 19; figures marked **[S]** must not be quoted externally without stating the range. Contains a correction: the 5–10% expansion joints share claim does not survive cross-checking.

11 — Why India's "Expansion Joints" Share and "Bellows" Share Differ

The question: `10-india-market-data.md` reports India at **5–10%** of the global expansion joints market but only **~2.7%** of the global bellows market. Why the difference?

Short answer: three separate things are going on, and one of them is a correction to what I told you earlier.

1. **They are different markets.** Expansion joints and bellows overlap but neither contains the other, so a share against one is not comparable to a share against the other.
2. **There is a genuine structural effect.** India really does index higher in heavy, fabricated, freight-protected products than in light, precision, globally-traded ones. That part of the story is real and strategically important.
3. **But the specific 5–10% figure does not survive cross-checking.** I should not have quoted it without testing the denominator. Working it through, India's expansion joints share is more likely **1–5%**, and the honest position is that the 5–10% claim is unverifiable. Details in section 3.

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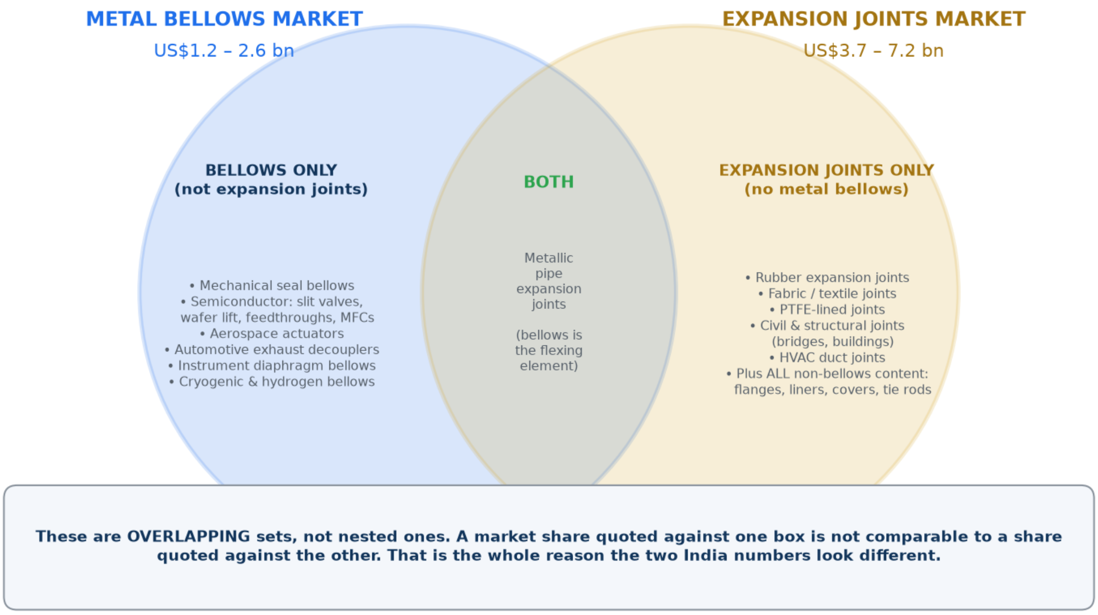
1. They are different markets, and neither is a subset of the other

Chart 17 — Why "bellows" and "expansion joints" are different markets
Note that the expansion joints market is measured 2-3x LARGER than the bellows market



Global expansion joints [S]: Dataintelo US$3.72bn (2024); Growth Market Reports US$3.87bn (2024); Marketintelo US$6.2bn (2025); Future Market Report US$6.2bn (2025). Metallic share of that: 38.5% (Marketintelo), 46.7% (Future Market Report), 64.7% (Dataintelo). Global metal bellows [S]: US$1.2-2.6bn — see chart 1. The expansion joints figure is larger because it includes non-metallic joints, civil/structural joints, and all the non-bellows hardware content of a finished joint.

An **expansion joint** is a finished piping component. A **bellows** is the flexible convoluted element inside it. So they overlap in metallic pipe expansion joints — but each market contains large areas the other does not:

Bellows but not expansion joints	Both	Expansion joints but not metal bellows
Mechanical seal bellows	**Metallic pipe expansion joints** — the bellows is the flexing element	Rubber expansion joints
Semiconductor: slit valves, wafer lift, feedthroughs, mass flow controllers		Fabric / textile joints (flue gas ducting)
Aerospace actuators, fuel systems		PTFE-lined joints
Automotive exhaust decouplers		Civil and structural joints (bridges, buildings)
Instrument diaphragm bellows		HVAC duct joints
Cryogenic and hydrogen bellows		**All the non-bellows content of a finished joint:** flanges, liners, covers, tie rods, engineering, hardware

The size relationship confirms this

Market	Global size	Sources
Metal bellows	**US$1.2 – 2.6 bn**	Four "all metal bellows" vendors, see `09` chart 1
Expansion joints	**US$3.7 – 7.2 bn**	Dataintelo US$3.72bn (2024); Growth Market Reports US$3.87bn (2024); Marketintelo US$6.2bn (2025); Future Market Report US$6.2bn (2025)
— of which metallic	38.5% – 64.7%	Marketintelo 38.5%; Future Market Report 46.7%; Dataintelo 64.7%

The expansion joints market is measured **2–3x larger** than the bellows market. That only makes sense once you see that it includes rubber, fabric, PTFE and civil joints — plus all the hardware value around the bellows — while excluding the many bellows applications that are not piping joints at all.

Practical consequence: you cannot compare a share figure from one market against a share figure from the other. They are answers to different questions.

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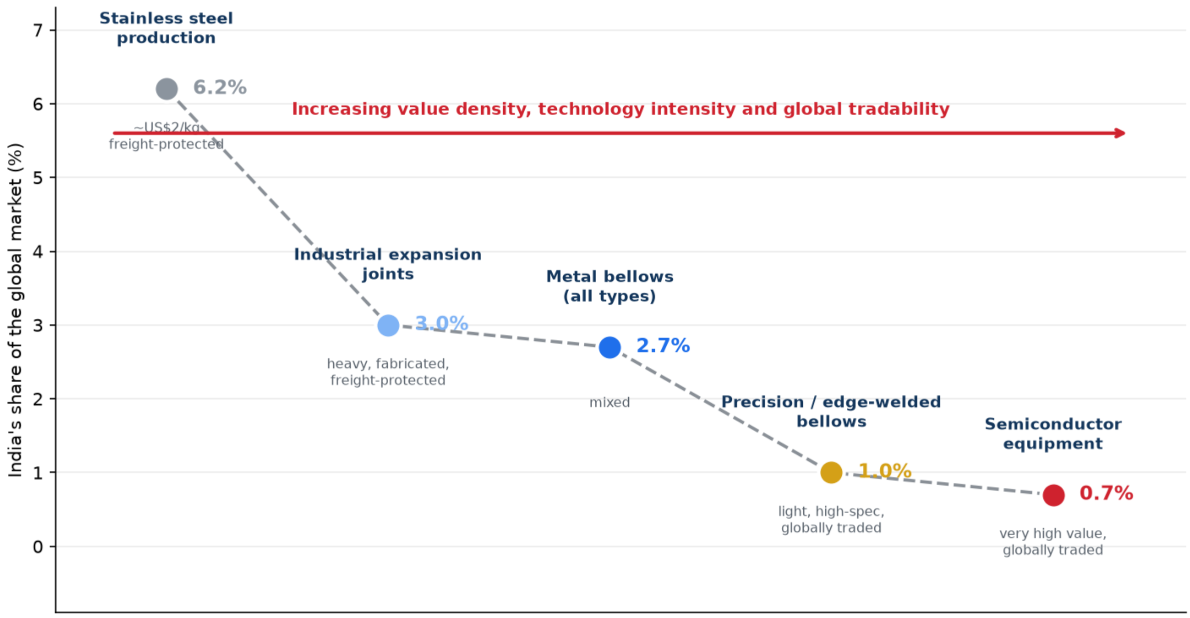
2. The real structural effect

Setting the measurement problem aside, there is a genuine reason India would index higher in expansion joints than in precision bellows, and it is worth understanding because it drives the strategy.

Chart 19 — The pattern is systematic, and it is the actual strategic finding
India's share collapses as products get lighter, more precise and more globally traded



Stainless production ~6.2% [C, 2021]. Expansion joints ~3.0% and bellows ~2.7% are our estimates. Precision/edge-welded ~1.0% and semiconductor equipment ~0.7% are modelled from the SEAJ 'Rest of World' bucket [P]. All points except the stainless figure are estimates. WHY IT HAPPENS: heavy fabricated products are freight- and tariff-protected, so every industrialising economy builds domestic capacity roughly in line with its industrial capex. Light, high-specification products are globally traded and concentrate in whoever holds the qualification history — the US, Japan, Korea, Germany. IMPLICATION: enter where India is strong (left of this chart) and use that position to earn the right to move right.

Market	India's estimated share	Product character
Stainless steel production	~6.2% (2021)	~US$2/kg, freight-protected
Industrial expansion joints	~3%	Heavy, fabricated, freight-protected
Metal bellows, all types	~2.7%	Mixed
Precision / edge-welded bellows	~1%	Light, high-specification, globally traded
Semiconductor equipment	~0.7%	Very high value, globally traded

Why the pattern exists. A 3-metre-diameter expansion joint for a cement plant cannot economically be shipped across the world — freight, duty and lead time all favour local supply. So every industrialising economy builds domestic expansion joint capacity roughly in proportion to its own industrial capex, and India has a lot of refineries, power plants, steel, cement and fertiliser. Freight and tariffs act as a protective moat.

A 30-gram edge-welded bellows for a slit valve has none of that protection. It is light, air-freightable, and its cost is dominated by qualification history rather than material or transport. So production concentrates wherever the qualification history sits — the US, Japan, Korea, Germany — and India's share collapses.

This is the actual strategic finding, and it is why the entry ladder in `07-research-reconciliation.md` starts at mechanical seals and industrial bellows: you enter on the left of that chart, where India is genuinely competitive, and use the position and reference base to earn the right to move right.

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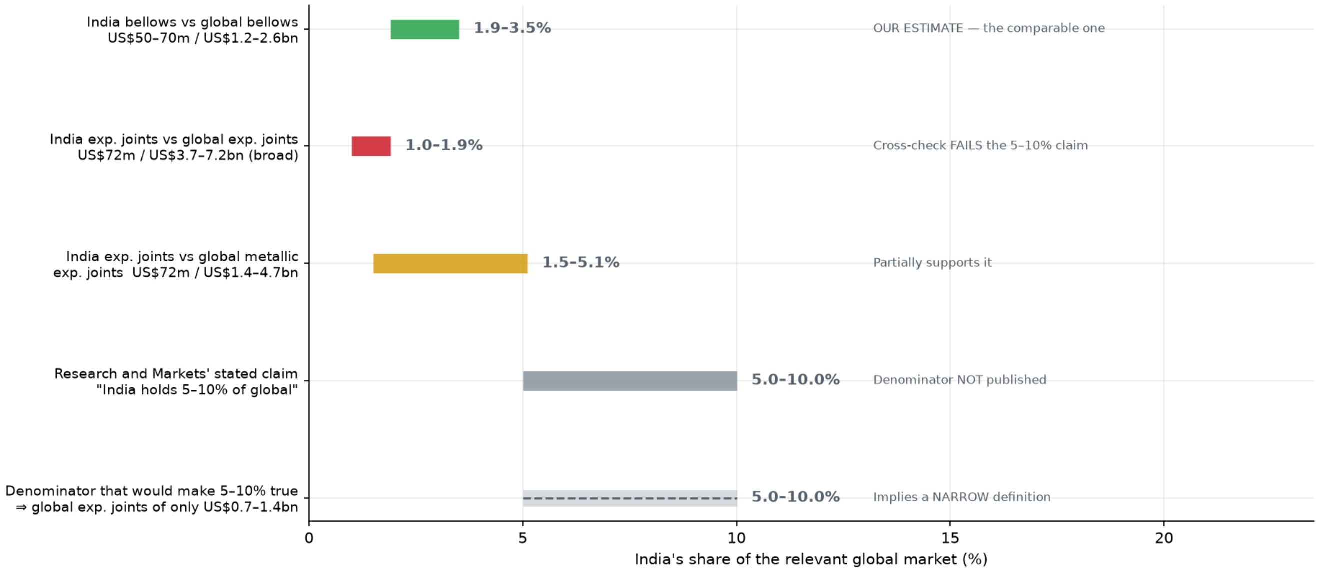
3. Correction: the 5–10% figure does not hold up

Research and Markets states two things on the same page:

- India's expansion joints market was **US$72.32 m in 2024**
- **"India currently holds 5%–10% share of the global expansion joints market"**

Put those together and they imply a global expansion joints market of only **US$0.72 – 1.45 bn**. But four other vendors put it at **US$3.7 – 7.2 bn**. Against that denominator, India at US$72m is **1.0% – 1.9%** — well below my bellows estimate of 2.7%, not above it.

Chart 18 — Same country, same year, five different answers
India's "share" ranges from 1.0% to 10% depending purely on how the denominator is drawn



THE KEY POINT: Research and Markets states both that India's expansion joints market was US$72.32m in 2024 AND that India holds 5-10% of the global expansion joints market. Those two statements together imply a global market of only US$0.72-1.45bn. Four other vendors put the global expansion joints market at US$3.7-7.2bn. So either R&M is using a much narrower definition (industrial metallic piping only, excluding civil, rubber and fabric joints), or the 5-10% figure does not hold. R&M does not publish its global denominator, so this cannot be resolved from public sources. RECOMMENDATION: use ~2-3% for bellows. Treat 5-10% as unverified and do not put it in a partner document.

Basis	India's share	Verdict
India bellows US$50–70m ÷ global bellows US$1.2–2.6bn	**1.9 – 3.5%**	Our estimate; internally consistent
India EJ US$72m ÷ global EJ US$3.7–7.2bn (broad)	**1.0 – 1.9%**	Cross-check **fails** the 5–10% claim
India EJ US$72m ÷ global **metallic** EJ US$1.4–4.7bn	**1.5 – 5.1%**	Partially supports it at the top of the range
R&M's stated claim	5 – 10%	Denominator not published
Denominator required for 5–10% to be true	⇒ global EJ of only US$0.7–1.4bn	Implies a much narrower definition

Two possible explanations, and I cannot distinguish them from public sources:

- **R&M uses a narrow definition.** Their India report segments only industrial applications — petrochemical, power, oil and gas, metals, cement, chemicals, industrial gases, railways, electrical — with no civil or structural category. If their global denominator is similarly restricted to industrial metallic piping joints, US$0.7–1.4bn is plausible and the 5–10% could be internally correct within their own frame.
- **The 5–10% figure is simply wrong,** or refers to something other than value share (units, or a specific sub-segment).

R&M does not publish the global denominator behind the claim, so this cannot be resolved without buying the report.

What to use instead

Figure	Status
India ≈ 2–3% of the global metal bellows market	**Use this.** Triangulated, internally consistent, and the relevant market for Iwatani
India ≈ 1–5% of the global expansion joints market	Use as a range if asked; note the denominator problem
"India holds 5–10% of the global expansion joints market"	**Do not put this in a partner or board document.** Unverified, and it fails cross-checking against four other vendors

I have corrected `10-india-market-data.md` accordingly.

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4. What actually changes as a result

Very little, and that is worth saying plainly.

Conclusion	Effect of this correction
India is a small share of the global bellows market (~2–3%)	**Unchanged.** This was always the load-bearing figure
India's absolute market is US$50–70m	**Unchanged.** Derived from company filings, not from any share claim
Iwatani's material slice in India is ~US$3–9m/yr	**Unchanged**
India punches above its weight in heavy industrial products and below it in precision	**Unchanged in direction, weaker in magnitude.** The gradient is real; it is shallower than 5–10% vs 2.7% implied
Entry ladder: mechanical seals → hydrogen/cryogenic → semiconductor	**Unchanged, and slightly reinforced.** If India's expansion joint share is 1–5% rather than 5–10%, there is more headroom in rung 1, not less
Do not anchor the business case on Indian domestic demand	**Unchanged**

The one thing that does change is a presentational risk that has now been removed. Had "India holds 5–10% of the global expansion joints market" gone into a document for Jindal or an Iwatani investment committee, a numerate reader could have dismantled it in one line by asking what the global market was. That is exactly the failure mode this pack's grading system exists to prevent, and it caught it one step late rather than not at all.

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5. The general lesson for this project

Every market share figure is a fraction, and **the denominator is where the deception lives.** Before quoting any share in this programme, ask three questions:

1. **What exactly is in the denominator?** Metallic only, or including rubber and fabric? Industrial only, or including civil and structural? Finished component value, or just the flexing element?
2. **Does the vendor publish the denominator, or only the share?** If only the share, treat it as unverified.
3. **Does it survive against an independent estimate of the same denominator?** If four other vendors put the global market 3–5x higher, the share claim is the thing that is wrong.

The bellows and expansion joint literature fails all three tests routinely. This is the third arithmetic inconsistency this project has caught — after the North American semiconductor bellows figure that exceeded half the global market, and the end-use shares that summed to 104%.

New sources introduced in this file

Source	URL	Grade
Dataintelo — Expansion Joints Market (US$3.72bn 2024)	https://dataintelo.com/report/expansion-joints-market	**[S]**
Dataintelo — Expansion Joint Market 2034 (metallic 64.7% share)	https://dataintelo.com/report/global-expansion-joint-market	**[S]**
Growth Market Reports — Expansion Joint Market (US$3.87bn 2024)	https://growthmarketreports.com/report/expansion-joint-market	**[S]**
Marketintelo — Expansion Joint Market (US$6.2bn 2025, metallic 38.5%)	https://marketintelo.com/report/expansion-joint-market	**[S]**
Future Market Report — Expansion Joint Market (US$6.2bn 2025, metal 46.7%)	https://www.futuremarketreport.com/industry-report/expansion-joint-market	**[S]**

Chart index

Chart	File	Purpose
17	`17-market-overlap.png`	Bellows and expansion joints as overlapping, not nested, markets
18	`18-india-share-denominators.png`	India's share under five different denominators
19	`19-share-vs-value-density.png`	The systematic decline in India's share as value density rises

Generated by `charts/make_reconciliation_charts.py`.